

Cosmic Binding Basins

A Dimensionless Boundary Index for Gravitational Membership in Λ CDM

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Abstract

We propose Cosmic Binding Basins (TEC-2), a phenomenological Λ CDM-compatible framework in which dark energy is treated as a large-scale boundary-setting term for gravitational membership, rather than as an internal dynamical disruption mechanism within already bound systems.

$$T\Lambda(R,M) = \Omega\Lambda H_0^2 R^3 / (GM)$$

This work is not claimed as a new physical law, but as a compact diagnostic derived from the Λ CDM turnaround scale.

In a pilot literature sample of 11 structures, all 9 central turnaround-like or zero-velocity estimates satisfy $T\Lambda \leq 1$, with median $T\Lambda = 0.319$. This is preliminary independent consistency, not proof of universality.

1. Physical Idea

TEC-2 does not introduce a new force. It reorganizes a known Λ CDM boundary scale into an operational index. The proposed publication-level statement is: dark energy is treated here as a large-scale boundary-setting term for gravitational membership, rather than as an internal dynamical disruption mechanism within already bound systems.

The testable question is whether observed boundaries such as zero-velocity or turnaround surfaces lie at or below $T\Lambda \approx 1$.

2. Formal Derivation

For a mass M enclosed within physical radius R , the characteristic inward gravitational acceleration is $a_g = GM/R^2$. In the low-redshift Λ CDM limit, the outward acceleration scale associated with dark energy is $a_\Lambda \approx \Omega\Lambda H_0^2 R$. Their ratio defines $T\Lambda$.

3. Relation To Existing Λ CDM Boundaries

For a cosmological constant, $T\Lambda = 1$ is equivalent to the maximum turnaround-radius scale in Schwarzschild-de Sitter/Kottler and spherical-collapse arguments.

$$R_{ta,max} = (3GM/\Lambda c^2)^{1/3} = [GM/(\Omega\Lambda H_0^2)]^{1/3}$$

4. Extensions: Dynamic Dark Energy, Environment And Geometry

$$TDE(R,M,z) = \{-[1 + 3w(z)]/2\} \Omega_{DE}(z) H(z)^2 R^3/(GM)$$

For $w = -1$ and $z = 0$, this reduces to $T\Lambda$. This version asks whether gravitational-basin boundaries evolve measurably with redshift.

TEC extension for dynamical dark energy

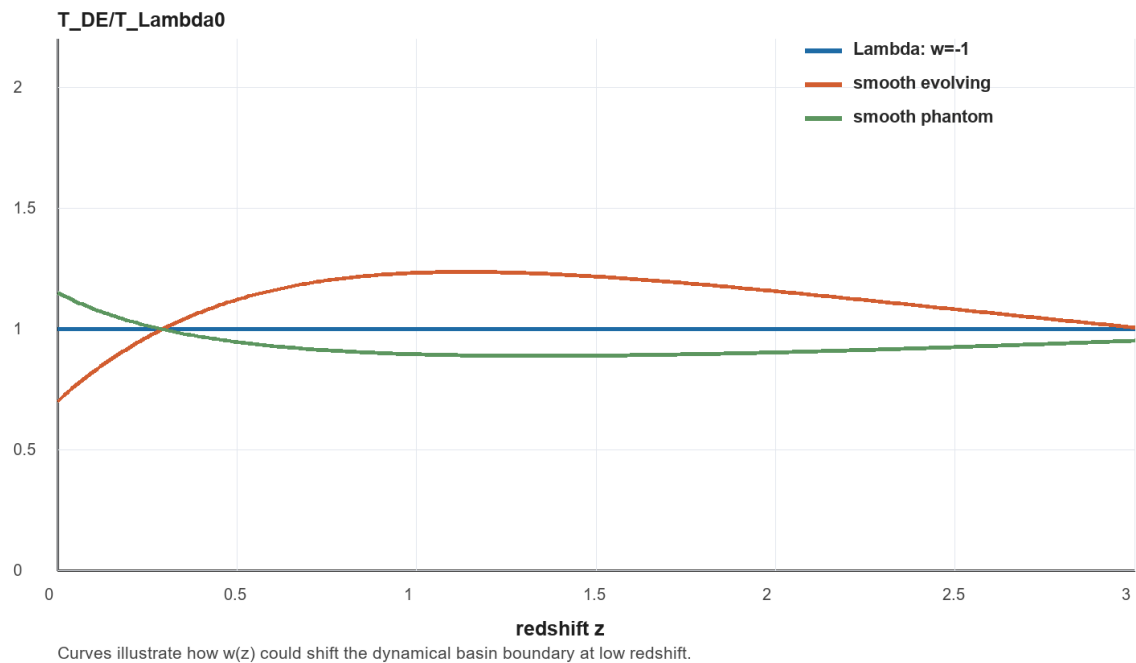


Figure 1. TEC extension for dynamical dark energy. Curves illustrate how different $w(z)$ histories could shift the boundary-setting term with redshift.

$$T_{\text{basin}}(R,M,z,\eta) = TDE(R,M_{\text{eff}},z) \cdot F_{\text{env}}(\eta)$$

The environmental correction is left as future work because F_{env} must be calibrated with simulations or survey data.

4.1 Geometric Basin Dominance (TEC-3 Proposal)

TEC-3 reframes basin membership through local Weyl or tidal curvature compared with the cosmological dark-energy curvature scale.

$$B\Lambda(x) = \sqrt{C\alpha\beta\gamma\delta \ C^{\alpha\beta\gamma\delta} / 48} / (\Lambda/3)$$

$$B\Lambda(R) = GM/(\Omega\Lambda \ H0^2 \ R^3) = 1/T\Lambda(R,M)$$

For simulations, a practical tidal proxy is $BE(x,z) = ||E_{ij}(x,z)|| / ADE(z)$, with $ADE(z) = -[1 + 3w(z)]\Omega DE(z)H(z)^2/2$. This is proposed, not validated here.

5. Pilot Data And Radius Definitions

The sample is intentionally small and heterogeneous. Radius type is listed because virial, zero-velocity, turnaround and core radii are not interchangeable.

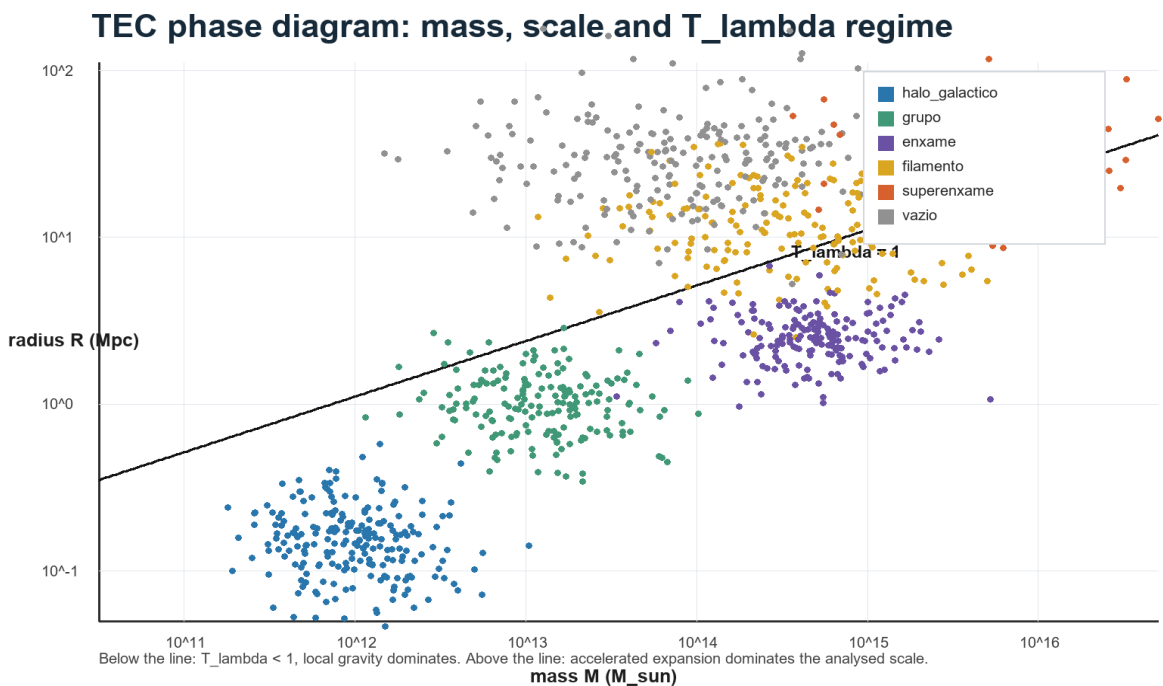
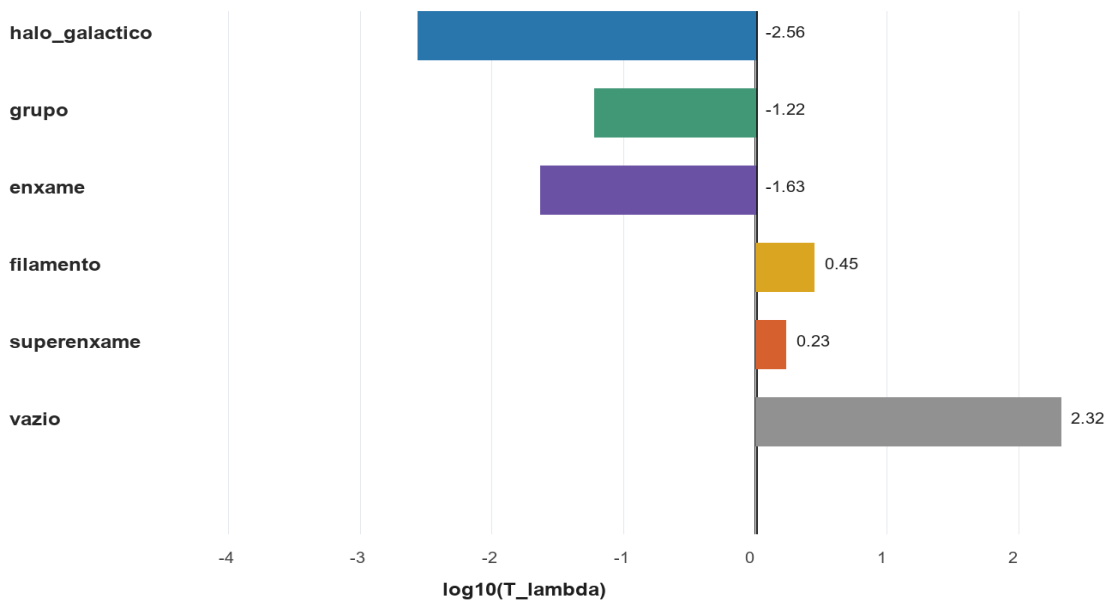


Figure 2. TEC phase diagram: mass, scale and T_λ regime in a synthetic proof-of-concept catalogue.

Median $\log_{10}(T_\lambda)$ by synthetic class



Negative values indicate bound-regime structures; positive values indicate expansion-dominated scales.

Figure 3. Median $\log_{10}(T_\lambda)$ by synthetic class.

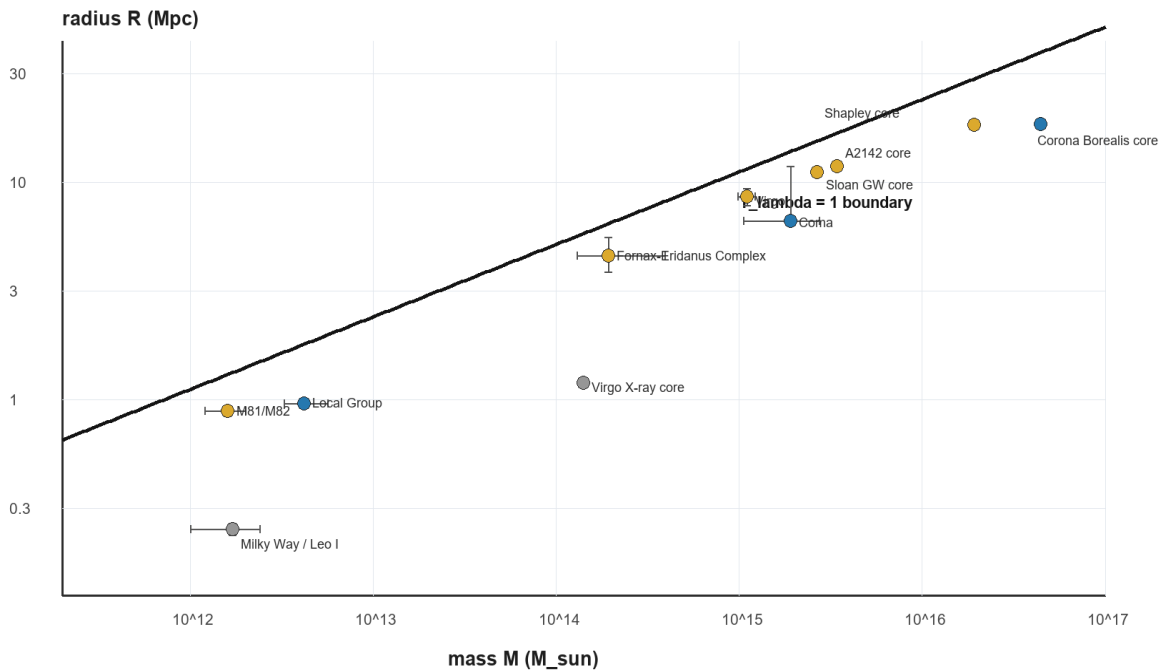
Object	Radius type	M [M_sun]	R [Mpc]	T_λ	Flag	Primary source
Milky Way / Leo I	bound satellite (not turnaround)	1.70e+12	0.25	0.007	no	Pavlidou & Tomaras 2014
Local Group	zero-velocity / local turnaround	4.17e+12	0.96	0.154	no	Karachentsev & Kashibadze 2006
M81/M82 Group	zero-velocity / local turnaround	1.60e+12	0.89	0.319	no	Karachentsev et al. 2002

Object	Radius type	M [M_sun]	R [Mpc]	TΛ	Flag	Primary source
Fornax-Eridanus Complex	zero-velocity / approximate turnaround	1.92e+14	4.60	0.367	no	Nasonova et al. 2011
Virgo Cluster	zero-velocity / local turnaround	1.10e+15	8.60	0.418	no	Karachentsev & Nasonova 2010
Virgo Cluster X-ray core	internal core radius (not turnaround)	1.40e+14	1.20	0.009	no	Urban et al. 2011
Coma Cluster	turnaround/infall bound; filamentary environment	1.90e+15	6.66	0.112	possible	Benisty et al. 2025 preprint
A2142 supercluster core	turnaround/collapsing core	3.41e+15	11.87	0.355	no	Einasto et al. 2018/2021
Sloan Great Wall core	turnaround/collapsing core	2.67e+15	11.13	0.373	no	Einasto et al. 2016/2021
Shapley supercluster core	turnaround/collapsing core	1.93e+16	18.40	0.234	no	Einasto et al. 2021
Corona Borealis core	turnaround/candidate bound supercluster	4.45e+16	18.55	0.104	no	Pearson et al. 2014

6. Results

All 9 central turnaround-like estimates satisfy $T\Lambda \leq 1$. The largest central value is $T\Lambda = 0.418$. Coma is the only object whose uncertainty envelope can approach or exceed the boundary.

TEC-2 pilot validation with published structures



All informative central estimates remain below $T_\Lambda = 1$; Coma can approach the limit under extreme uncertainty choices.

This is pilot consistency evidence for the TEC-2 boundary, not final proof of universality.

Figure 4. Pilot validation in the mass-radius plane. The solid black line is the $T\Lambda = 1$ boundary, i.e. $R = R_\Lambda(M)$.

Synthetic test: recovering the $R \sim M^{(1/3)}$ scaling

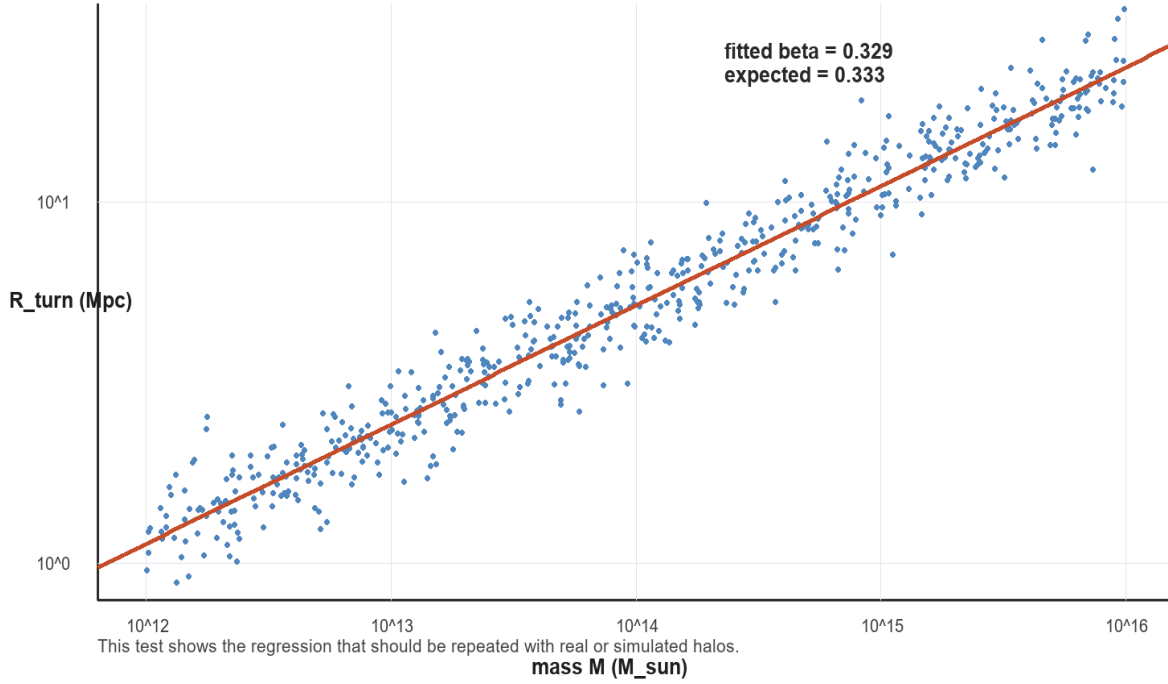


Figure 5. Synthetic sanity check recovering the expected $R \propto M^{(1/3)}$ scaling.

7. Discussion And Conclusion

The pilot result supports a cautious claim: structures already interpreted as bound, in turnaround, or in collapse are consistent with the TEC-2 boundary criterion. The result is not yet a discovery of new physics. It is a compact re-expression of a known Λ CDM scale that appears useful as a gravitational-basin classifier.

The next decisive test is comparative: apply TA and the proposed TEC-3 tidal proxy to a larger homogeneous simulation or survey sample, then test whether they improve boundary classification relative to mass, density, virial radius, turnaround radius and cosmic-web classifiers.

References

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